a Study design
Open data (no IRB)
Analyses (D = discovery, C = confirmatory)
Tahoe-100M single-cell drug atlas
379 drugs · 50 cancer cell lines (15 lung)
3 doses · 24 h · pseudobulk differential expression
Clinical cardiotoxicity labels
FDA DICTrank · FDA FAERS (openFDA)
Cell-line drug sensitivity and baseline state
PRISM · GDSC · CCLE RNA-seq,
mass-spectrometry proteomics, RPPA
D2 Reproducibility and mechanism of action (positive controls)
D1 · C1 Clinical cardiotoxicity: Tahoe response vs chemistry
D3 Pooled 24-h cell counts as a viability readout
D4 · C2 Response magnitude and proliferation program vs sensitivity
C3 Tahoe response vs baseline RNA vs baseline protein (MS, RPPA)
Pre-registered plan · permutation nulls · robustness · dated amendments
b
Drugs per analysis
D1 DICTrank most vs no
D1 DICTrank most+less vs no
C1 FAERS signal
D4 PRISM
D4 GDSC
C2 PRISM
C2 GDSC
C3 PRISM
C3 GDSC
D2 mechanism classes
100
167
128
51
43
44
37
43
37
62
Number of drugs
c
Signature reproducibility
Different drugs (median 0.22)
Same drug (median 0.68)
Density
Correlation of mid- vs high-dose signatures
d
Mechanism recognition
Permutation null
Observed 0.32 (p = 0.0005)
Chance 0.09
Permutations
Balanced accuracy
(1-nearest neighbour, 11 classes)
e
Correct by mechanism class
DNA synth./repair
EGFR/ERBB
Multi-TK
Other TK
Microtubule
CDK
HDAC
JAK/STAT
MEK
RAF
Hedgehog
6/17
1/8
1/7
0/7
1/5
0/3
2/3
2/3
3/3
1/3
0/3
Drugs in class
Correctly recognised
Drugs
f
Mean signature similarity between mechanism classes
Mean Pearson r
Within-class (diagonal) mean r = 0.28 · between-class mean r = 0.17